SIES GRADUATE SCHOOL OF TECHNOLOGY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

BE VII Semester

Mini Project Seminar

“Drug Interaction and Adverse-Effect Predictor”

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1. INTRODUCTION

THE CHALLENGE

Medication combinations can create interactions or change adverse-effect risk. Manually cross-checking fragmented sources is slow, and missing database records do not prove safety.

PROJECT OBJECTIVES

- Merge public biomedical data
- Predict interaction probability, type and side effects
- Rank concerns and show supporting paths
- Provide an inspectable web interface

1,706
Drug nodes in the drug universe

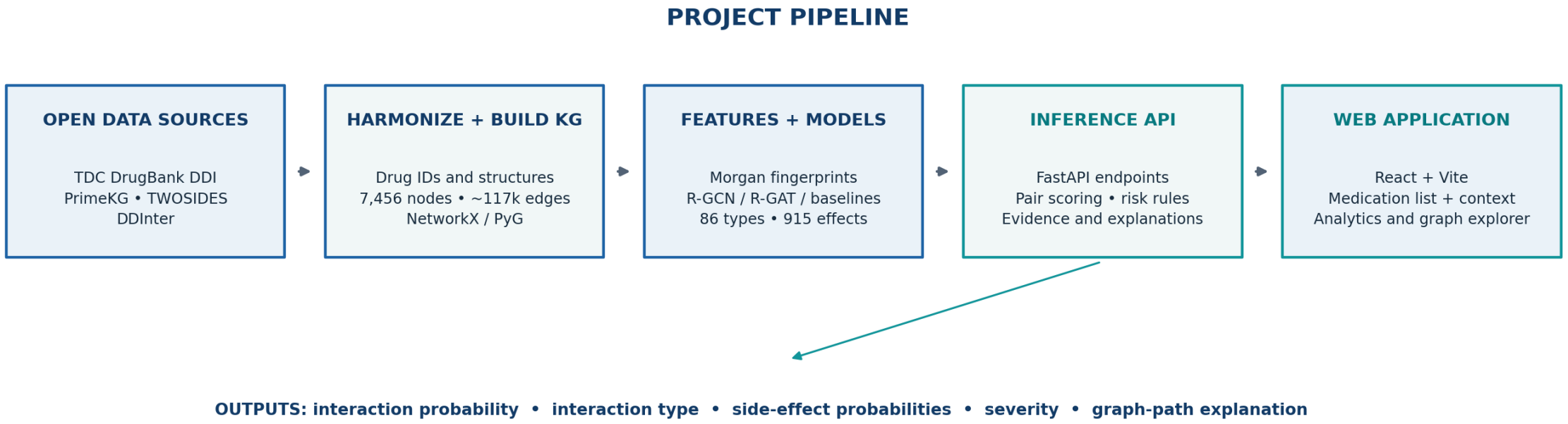
526,105
Unique known interaction pairs

7,456
Knowledge-graph nodes

915
Side-effect labels retained

Intended use: research and educational decision support only — not a medical device.

2. METHODOLOGY: END-TO-END PIPELINE



Key design decision: interaction labels are kept separate from graph relations to reduce label leakage into message passing.

2. METHODOLOGY: MODEL DESIGN

FINGERPRINT BASELINES

Morgan fingerprints

1,024 bits per drug

Logistic regression (SGD)

XGBoost

GRAPH MODELS

R-GCN: relational message passing

R-GAT: attention over relations

R-GCN + GIN: molecular graph encoder (not served)

THREE PREDICTION HEADS

1. Interaction probability
2. One of 86 interaction types
3. Multi-label probabilities for 915 side effects

EVALUATION SPLITS

- Random split: interacting pairs divided 80/10/10.
- Cold-start: test pairs include at least one drug unseen in training.

LABEL HANDLING

- Unrecorded pairs are unlabelled, not safe negatives.
- Metrics also use degree-matched unlabelled pairs as a harder set.

3. IMPLEMENTATION: APPLICATION WORKFLOW

1 ENTER

Search and select 2–20 medications; optional prescription text extraction.

2 ANALYSE

Score all drug pairs and apply severity and patient-context rules.

3 REVIEW

See regimen risk, flagged pairs, effects, evidence and severity source.

4 EXPLAIN

Open the pair path, model-attribution view, evidence and drug cards.

Other modules: patient-context alerts • knowledge-graph explorer • model analytics • RxNorm normalization • FAERS evidence • safer-alternative suggestions • audit log

Backend: FastAPI | Frontend: React + Vite | Graph/model layer: PyTorch Geometric + NetworkX

4. RESULTS: EVALUATION DESIGN

DATA SCALE

1,706 drugs

1,454,365 possible pairs

526,105 unique known pairs

~117k biomedical graph edges

METRICS

AUROC and AUPRC

F1 with threshold tuned on validation

Hits@20 / Hits@50 / Hits@100

Auxiliary type and side-effect scores

ROBUSTNESS CHECK

Random split is easier and may be optimistic.

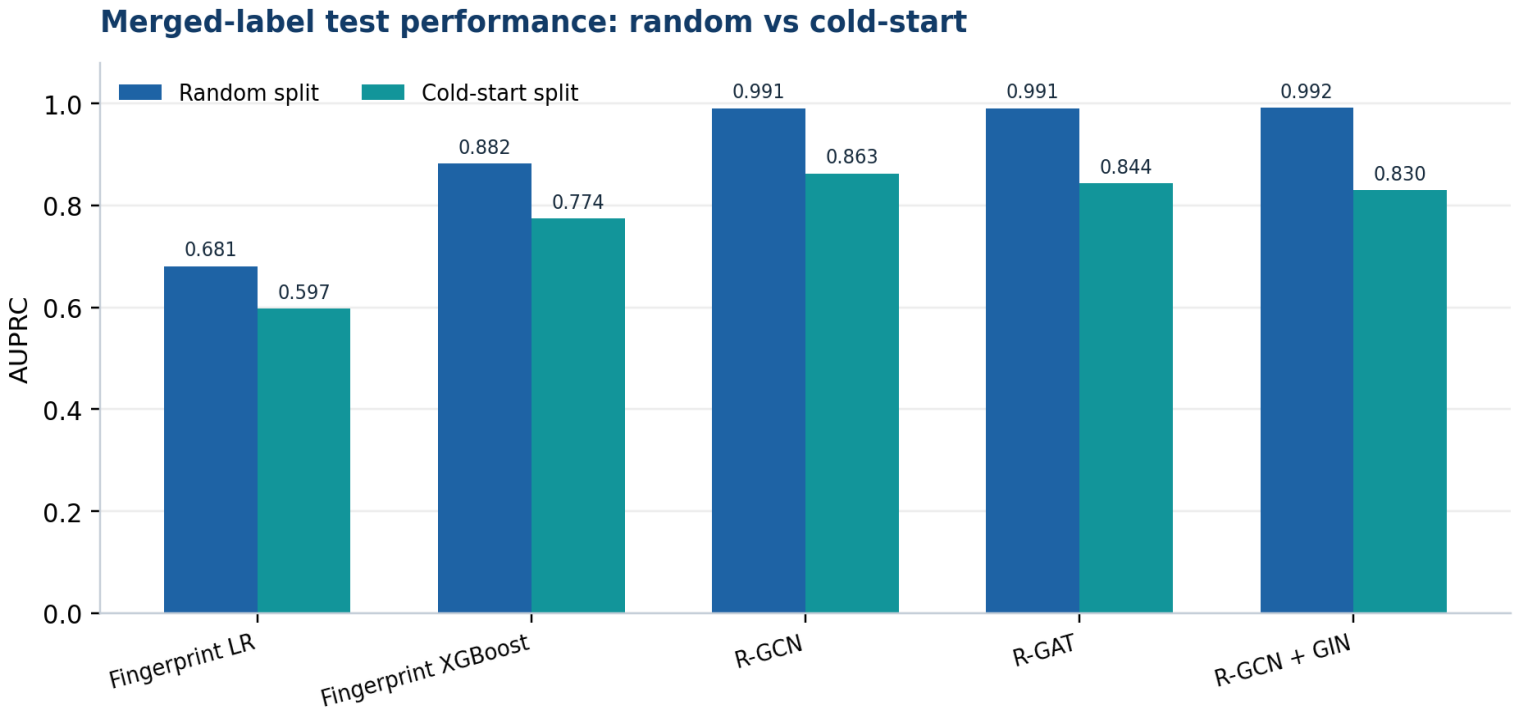
Cold-start tests drugs held out from training.

Degree-matched unlabelled pairs add difficulty.

Evaluation uses unlabelled pairs as comparison examples — they may include interactions absent from the source databases.

- Therefore, metrics are lower-bound-style measures of ranking against recorded labels, not estimates of clinical safety.
- The model card explicitly states that the system has no prospective clinical validation.

4. RESULTS: INTERACTION PREDICTION



BEST SERVED MODEL

R-GCN

Cold-start AUROC 0.870

Cold-start AUPRC 0.863

WHY COLD-START?

Includes drug pairs with at least one drug not seen during training.

Random-split R-GCN: AUROC 0.992 / AUPRC 0.991 • Degree-matched cold-start: AUROC 0.843 / AUPRC 0.837

4. RESULTS: EXPLAINABILITY AND LIMITATIONS

ILLUSTRATIVE CASE STUDY

Warfarin + Ibuprofen

Model score: 0.76

Top model-attribution neighbour: CYP2C9

Highest ranked path node: gastrointestinal hemorrhage

EXPLANATION AGREEMENT

In 10 curated pairs, the model's top neighbour was on a KG path in 80%.

Across 200 test pairs, the top-5 overlap was 11%; explanations are complementary, not identical.

IMPORTANT LIMITATIONS

- No prospective clinical validation; statistical predictions can be incomplete or wrong.
- No flag does not establish safety. Confirm with an authoritative drug reference and healthcare professional.
- Severity for pairs without DDInter annotation is a project type-based estimate; evidence sources may be biased or stale.

5. CONCLUSION AND FUTURE WORK

CONCLUSION

The project integrates public biomedical data, a knowledge graph, relational GNN predictions, interaction severity, side-effect estimates and explanation paths in a single web application. The R-GCN is the strongest served model among the evaluated options for cold-start interaction ranking.

FUTURE WORK

Temporal and external validation; better-calibrated and less imbalanced side-effect prediction; expanded drug coverage; clinician review of explanations; clearer evidence freshness and uncertainty indicators.

Takeaway: a promising research prototype for review and exploration — not a stand-alone prescribing or medication-safety authority.

6. REFERENCES

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Thank you